Abel Haro Armero

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Professional Summary

Computer Engineer with a Master's Degree in Computer Engineering and Networks and a Bachelor's Degree in Informatics Engineering from Universitat Politècnica de València. I focus on backend development, cloud infrastructure, and maintainable system design. I enjoy turning real product needs into robust, well-engineered solutions while continuing to learn and explore new technologies.

Education

- **Master's Degree in Computer and Network Engineering** *september 2025 - july 2026*
  Universitat Politècnica de València average grade 8.9
- **Computer Engineering** *september 2021 - july 2025*
  Universitat Politècnica de València average grade 8.6

Languages

Spanish - Native
English - B2

Professional Experience

- **Internship at the DISCA Department, UPV** *october 2024 - july 2025*
  Universitat Politècnica de València
  – Developed a computer vision system to detect defects in industrial objects from images, covering data preparation, model training and result validation with Python and YOLO.
  – Trained and evaluated models with Ultralytics, tuning the inference workflow for NVIDIA Jetson hardware with TensorRT and deployment in low-power environments.

- **Internship at SOLTECSIS S.L.** *july 2024*
  SOLTECSIS S.L.
  – Debugged and fixed issues during the migration of the open-source FWCloud project from JavaScript to TypeScript, working on existing modules and reviewing behavior affected by typing.
  – Collaborated on codebase refactoring to improve maintainability, readability and type safety, following the project structure and team reviews.

Personal and Student projects

- **URL Shortener** *March 2026*
  github.com/AbelHaro/url-shortener url-shortener.abelharo.me
  – Technologies: React, Go, PostgreSQL, Docker, Google Cloud.
  – Developed a full-stack application to create, store and manage shortened URLs, designing the user flow and communication between frontend, API and database.
  – Implemented the React frontend, Go backend, PostgreSQL persistence, Docker containerization and Google Cloud deployment, prioritizing a simple and maintainable architecture.

- **Tracker - Object Tracking Library** *Under development*
  github.com/AbelHaro/tracker
  – Technologies: C++23, Python, ByteTrack, YOLO, OpenCV.
  – Developing an object tracking library in C++23 with Python bindings and a ByteTrack implementation to associate detections across frames and maintain an ID for each object.
  – Includes an example integrating YOLO and OpenCV for object detection and tracking in video.

- **Defect detection in objects using convolutional neural networks** *October 2024 - June 2025*
  github.com/AbelHaro/TFG | Bachelor's thesis
  – Technologies: Python, Computer Vision, CNN, YOLO, Ultralytics, TensorRT, NVIDIA Jetson.
  – Developed a real-time defect detection system for industrial product images, comparing YOLO models and optimizing inference on low-power hardware.